RELATIVELY PERFECT COMPLEXES WITH VANISHING NEGATIVE EXTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition spaces-more-morphisms-definition-relatively-perfect (canonical tag 0DKN). cited at statement line 5329. S-perfect complexes.
- Lemma quot-lemma-complexes-open-neg-exts-vanishing (canonical tag 0DLC). cited at proof line 5350. Collected card: open vanishing lemma.

Source reference. Original definition 0DKN: sources/stacks-chow/tags.txt, source lines 17018–17018. Cited at statement line 5329: definition of S-perfect complexes.

Source reference. Original lemma 0DLC: sources/stacks-chow/tags.txt, source lines 17042–17042. Cited at statement line 5350: the open vanishing lemma being specialized.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is proper, flat, and of finite presentation. Let $E \in D(\mathcal{O}_X)$. Assume*

(1) *E is S-perfect (More on Morphisms of Spaces, Definition 0DKN), and*
(2) *for every point $s \in S$ we have*

$$\mathrm{Ext}^i_{\mathcal{O}_{X_s}}(E_s, E_s) = 0 \quad \text{for} \quad i < 0$$

where E_s is the pullback to the fibre X_s.

Then

(a) *(1) and (2) are preserved by arbitrary base change $V \to Y$,*
(b) *$\mathrm{Ext}^i_{\mathcal{O}_{X_V}}(E_V, E_V) = 0$ for $i < 0$ and all V over Y,*
(c) *$V \mapsto \mathrm{Hom}_{\mathcal{O}_{X_V}}(E_V, E_V)$ is representable by an algebraic space affine and of finite presentation over Y.*

Here X_V is the base change of X and E_V is the derived pullback of E to X_V.

Proof. Immediate consequence of Lemma 0DLC. □

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